

Dendritic cells: sentinels against pathogens.

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Dendritic cells (DCs) are the most potent antigen-presenting cells, and are regarded as "natural adjuvants" for the induction of primary T or T-dependent immunity. DCs in the peripheral sites capture and process antigens. Encounter of exogenous or endogenous stimuli mature the function of DCs, and they thus acquire T-cell stimulatory capacity and distinct chemotactic behavior which enables them to migrate to lymphoid tissue. In the secondary lymphoid organs, they present antigens to T- and B-cells and stimulate their proliferation. Dendritic cells are also involved in tolerance induction, in particular, to self antigens. DCs also play a key role in the transmission of many pathogens, and therefore may become targets for designing new therapies. DCs have been manipulated in vitro and in vivo for cancer immunotherapy. In this article, we provide a concise overview of DC biology and its current and future role in clinical settings.

Dendritic cells and monocytes as accessory cells in T-cell responses in man. II. Function as antigen-presenting cells.

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The main antigen-presenting cells (APC) of human blood are reported to be the dendritic cells (DC), whereas monocytes (Mo) are only weakly or not at all capable of inducing T-cell immune responses to the soluble antigen purified protein derivative (PPD). In contrast, we found Mo to have a suppressive effect on the APC function of DC in vitro. Removal of Mo by adherence resulted in an increased APC activity, even though Mo produce more interleukin 1 (Il-1) than DC. Furthermore, addition of the prostaglandin synthesis inhibitor indomethacin gave rise to increased Il-1 production, HLA-class II expression, and a stronger antigen-specific T-cell response to PPD. Taken together, our studies indicate that the superior accessory cell function of DC compared with Mo in in vitro cultures may, at least partly, be attributed to the prostaglandin E2 production by Mo and more stable expression of HLA-class II molecules on DC.

Enzymatic reactions of *Clostridium difficile* in aerobic and anaerobic environments with the RapID-ANA II identification system.

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The RapID-ANA II anaerobic identification system (Innovative Diagnostic Systems, Inc., Atlanta, Ga.) was used to determine whether the incubation environment affects enzyme detection. Twenty strains of *Clostridium difficile* were tested in aerobic, anaerobic, and low-CO₂ anaerobic incubation environments. The percentages of enzymes detected in reactions with the following substrates were noted in the three incubation environments: phenylalanine-beta-naphthylamide, aerobic, 0%; anaerobic, 35%; low-CO₂ anaerobic, 35%; arginine-beta-naphthylamide, aerobic, 5%; anaerobic, 55%; low-CO₂ anaerobic, 75%; pyrrolidonyl-beta-naphthylamide, aerobic, 5%; anaerobic, 65%; low-CO₂ anaerobic, 65%. When the aerobic incubation environment was compared with either the anaerobic or the low-CO₂ anaerobic incubation environments, the results were statistically different with respect to enzyme detection in reactions with the substrates listed above. The results for the anaerobic and low-CO₂ anaerobic environments were not statistically different. The study was repeated twice. Statistical comparisons

between the three environments were consistent with the results presented above, with the following exceptions. The aerobic and the anaerobic environments were not different in a reaction with phenylalanine-beta-naphthylamide in one of the runs, and there was no significant difference between the three environments in a reaction with arginine-beta-naphthylamide in another run. These results suggest that some of the enzymes used in the identification of clinical anaerobes appear to be inactive in an environment containing oxygen.

Tracking gene expression and oxidative damage of O₂-stressed *Clostridioides difficile* by a multi-omics approach.

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Clostridioides difficile is the major pathogen causing diarrhea following antibiotic treatment. It is considered to be a strictly anaerobic bacterium, however, previous studies have shown a certain and strain-dependent oxygen tolerance. In this study, the model strain *C. difficile* 630Δerm was shifted to micro-aerobiosis and was found to stay growing to the same extent as anaerobically growing cells with only few changes in the metabolite pattern. However, an extensive change in gene expression was determined by RNA-Seq. The most striking adaptation strategies involve a change in the reductive fermentation pathways of the amino acids proline, glycine and leucine. But also a far-reaching restructuring in the carbohydrate metabolism was detected with changes in the phosphotransferase system (PTS) facilitated uptake of sugars and a repression of enzymes of glycolysis and butyrate fermentation. Furthermore, a temporary induction in the synthesis of cofactor riboflavin was detected possibly due to an increased demand for flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD) in redox reactions. However, biosynthesis of the cofactors thiamin pyrophosphate and cobalamin were repressed deducing oxidation-prone enzymes and intermediates in these pathways. Micro-aerobically shocked cells were characterized by an increased demand for cysteine and a thiol redox proteomics approach revealed a dramatic increase in the oxidative state of cysteine in more than 800 peptides after 15 min of micro-aerobic shock. This provides not only a catalogue of oxidation-prone cysteine residues in the *C. difficile* proteome but also puts the amino acid cysteine into a key position in the oxidative stress response. Our study suggests that tolerance of *C. difficile* towards O₂ is based on a complex and far-reaching adjustment of global gene expression which leads to only a slight change in phenotype.

Clostridioides difficile infection: an update.

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Clostridioides difficile (*C. difficile*) is a Gram-positive, spore-forming anaerobic bacterium emerged as a leading cause of diarrhea globally. CDI's (*Clostridioides difficile* infection) impact on healthcare systems is concerning due to high treatment cost and increased hospitalisation time. The incidence of CDI has been influenced by hypervirulent strains such as the 027 ribotype, responsible for significant outbreaks in North America and Europe. CDI's epidemiology has evolved, showing increased community-acquired cases alongside traditional hospital-acquired infections. Mortality rates remain high, with recurrent infections further elevating the risk. Transmission of *C. difficile* primarily occurs via spores, which survive in healthcare settings and play a pivotal role in transmission. Not only health workers, but also the food

chain could have a significant impact on the transmission of infection, although no confirmed foodborne cases have been documented. Pathogenicity of *C. difficile* involves spore germination and toxin production. Toxins A and B can cause cellular damage and inflammatory responses in the host, leading to colitis. Clinical picture can range from mild diarrhea to fulminant colitis with toxic megacolon, and bowel perforation. Risk factors for CDI include antibiotic exposure, advanced age, hospitalization, and use of proton pump inhibitors. Patients who experience abdominal surgery or patients with inflammatory bowel disease (IBD) are particularly susceptible due to their compromised gut microbiota. Management of CDI has evolved, with fidaxomicin emerging as a superior treatment option over vancomycin for initial and recurrent infections due to its reduction of recurrence rate. Faecal microbiota transplantation (FMT) is effective for recurrent CDI, restoring gut eubiosis. Bezlotoxumab, a monoclonal antibody against *C. difficile* toxin B, has shown promise in reducing recurrence rates. Severe cases of CDI may require surgical intervention, particularly in instances of toxic megacolon or bowel perforation. In conclusion, CDI remains a significant clinical entity. Further research are needed to improve patients' outcome and reduce the burden on healthcare systems.

Genetic mechanisms governing sporulation initiation in *Clostridioides difficile*.

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As an anaerobe, *Clostridioides difficile* relies on the formation of a dormant spore for survival outside of the mammalian host's gastrointestinal tract. The spore is recalcitrant to desiccation, numerous disinfectants, UV light, and antibiotics, permitting long-term survival against environmental insults and efficient transmission from host to host. Although the morphological stages of spore formation are similar between *C. difficile* and other well-studied endospore-forming bacteria, the *C. difficile* genome does not appear to encode many of the known, conserved regulatory factors that are necessary to initiate sporulation in other spore-forming bacteria. The absence of early sporulation-specific orthologs suggests that *C. difficile* has evolved to control sporulation initiation in response to its unique and specific ecological niche and environmental cues within the host. Here, we review our current understanding and highlight the recent discoveries that have begun to unravel the regulatory pathways and molecular mechanisms by which *C. difficile* induces spore formation.

Clostridioides difficile Biology: Sporulation, Germination, and Corresponding Therapies for *C. difficile* Infection.

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Clostridioides difficile is a Gram-positive, spore-forming, toxin-producing anaerobe, and an important nosocomial pathogen. Due to the strictly anaerobic nature of the vegetative form, spores are the main morphotype of infection and transmission of the disease. Spore formation and their subsequent germination play critical roles in *C. difficile* infection (CDI) progress. Under suitable conditions, *C. difficile* spores will germinate and outgrow to produce the pathogenic vegetative form. During CDI, *C. difficile* produces toxins (TcdA and TcdB) that are required to initiate the disease. Meanwhile, it also produces spores that are responsible for the persistence and recurrence of *C. difficile* in patients. Recent studies have shed light on the regulatory mechanisms of *C. difficile* sporulation and germination. This review is to summarize recent advances on the regulation of sporulation/germination in *C. difficile* and the

corresponding therapeutic strategies that are aimed at these important processes.

Laboratory maintenance of *Clostridium difficile*.

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Clostridium difficile is a Gram-positive, spore-forming, anaerobe and is the leading cause of antibiotic-associated diarrhea, pseudomembranous colitis, and toxic megacolon. Essential to the lifestyle of *C. difficile* is the ability to form a metabolically dormant spore, germinate, and grow out upon appropriate signals and elicit disease with the secretion of two toxins. To aid in the study of this organism, this unit describes the growth and maintenance of *C. difficile*. Included are methods to isolate *C. difficile* from environmental samples, grow in laboratory medium, and produce and purify spores.

.*Clostridioides difficile*.

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Clostridioides difficile is a spore-forming, anaerobic, intestinal pathogen that causes severe diarrhea that can lead to death. In 2011, *C. difficile* infected ~500000 people in the USA and killed ~29000 people. *C. difficile* infection (CDI) is the most common healthcare-related infection in the USA, leading to increased healthcare costs of \$4.8 billion. This pathogen transmits via the oral-fecal route as a highly contagious and resilient spore. Upon exposure to primary bile acids in the intestine, *C. difficile* germinates, and in the absence of colonization resistance from the normal microbiota, the bacterium colonizes the colon and produces toxins. These toxins inhibit actin polymerization in host cells, leading to cell death. *C. difficile* cells can then sporulate in the intestine and exit the body via diarrheal shedding. In culture, sporulation is induced at stationary phase in a nutrient-limiting environment, but the intestinal triggers of sporulation are still unknown.

Clostridioides difficile spore: coat assembly and formation.

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Clostridioides difficile (*C. difficile*) is a Gram-positive, spore-forming, toxin-producing, obligate anaerobic bacterium. *C. difficile* infection (CDI) is the leading cause of healthcare-associated infective diarrhoea. The infection is mediated by the spore, a metabolically inactive form of *C. difficile*. The spore coat acts as a physical barrier to defend against chemical insults from hosts and natural environments. The composition of spore coat has already been revealed; therefore, the interactive networks of spore coat proteins and the dynamic process of coat assembly are the keys to design strategies to control and cure CDI. This review gives a brief discussion of the signal processing and transcriptional regulation of *C. difficile* sporulation initiation. Following the discussion, the spore formation is also introduced. Finally, this review mainly focuses on the spore coat assembly, a poorly understood process in *C. difficile*, and important proteins that have been studied.
